

ROUTE B, CLARIFIED AGAINST THE NEAR-SERRIN PICTURE

WORKING NOTE (PLAN 3)

ABSTRACT. You proposed, for the graph route (Strategy A), to take the small problematic part of a near-Serrin boundary and *smooth it out* by passing to an interior level set, using the Poisson kernel (interior elliptic regularity), and then to bootstrap higher derivatives from the fact that $|\nabla u|$ varies little. This note explains, in full, what Route B does with the *same* input ($|\nabla u|$ close to constant on a level, in L^2), why it is the *opposite* design choice, and exactly where each route's single hard point sits. Everything is written out: the level-set defect, the Markov selection, the Rellich–Pohozaev and Weinberger identities, the trace-free Hessian rigidity, the small-gradient excision, the Serrin dictionary, and a worked ball-plus-tentacle illustration.

1. SETUP AND THE COMMON INPUT

Let $\Omega \subset \mathbb{R}^n$ be open, $|\Omega| = \omega_n$, with torsion function $-\Delta u = 1$ in Ω , $u = 0$ on $\partial\Omega$, $u > 0$ inside. Write $T(\Omega) = \int_{\Omega} |\nabla u|^2 = \int_{\Omega} u$, let B be the volume-matched ball, and set the scale-invariant Saint–Venant deficit and the Fraenkel asymmetry

$$\delta_T(\Omega) = \frac{T(B) - T(\Omega)}{T(B)} \geq 0, \quad \mathcal{A}(\Omega) = \min_{x_0} \frac{|\Omega \triangle B(x_0)|}{|\Omega|}.$$

The target is the sharp inequality $\mathcal{A}(\Omega)^2 \leq C(n) \delta_T(\Omega)$.

1.1. The level-set defect. For a regular value t write $E_t = \{u > t\}$, $\Sigma_t = \{u = t\}$, $m(t) = |E_t|$, $P(t) = \mathcal{H}^{n-1}(\Sigma_t)$, and the mean $\bar{f}_t = m(t)/P(t)$. From $-\Delta u = 1$ one has the two elementary identities

$$m(t) = \int_{E_t} (-\Delta u) = \int_{\Sigma_t} |\nabla u| d\mathcal{H}^{n-1}, \quad -m'(t) = \int_{\Sigma_t} \frac{d\mathcal{H}^{n-1}}{|\nabla u|}.$$

Cauchy–Schwarz, $P(t)^2 \leq (\int_{\Sigma_t} |\nabla u|^{-1}) (\int_{\Sigma_t} |\nabla u|)$, defines the *boundary-gradient defect*

$$D_H(t) := \left(\int_{\Sigma_t} \frac{d\mathcal{H}^{n-1}}{|\nabla u|} \right) \left(\int_{\Sigma_t} |\nabla u| d\mathcal{H}^{n-1} \right) - P(t)^2 \geq 0,$$

which vanishes exactly when $|\nabla u|$ is constant on Σ_t (the Serrin overdetermined condition). Expanding the square gives the exact *weighted-variance identity*

$$(1) \quad \int_{\Sigma_t} \frac{(|\nabla u| - \bar{f}_t)^2}{|\nabla u|} d\mathcal{H}^{n-1} = \frac{m(t)}{P(t)^2} D_H(t).$$

Thus $D_H(t)$ is precisely the variance of $|\nabla u|$ on Σ_t *weighted by* $1/|\nabla u|$; this weight is the protagonist throughout. The level-set deficit identity (Plan 2) reads $\delta_T(\Omega) \simeq_n \int (D_H(t) + D_I(t)) w(t) dt$ with a positive weight w and the isoperimetric defect $D_I \geq 0$; only D_H is used here.

1.2. The Markov selection.

Lemma 1.1. *Fix a small absolute constant $\theta \in (0, 1)$. There is a level $\hat{t}$ with $D_H(\hat{t}) \leq n^2 \omega_n \theta$ (equivalently, the left side of (1) at $\hat{t}$ is $\leq \theta$), with the discarded layer $|\Omega \setminus E_{\hat{t}}| = O_\theta(\delta_T(\Omega))$, and with $\hat{t}$ a regular value.*

Proof sketch. Drop $D_I \geq 0$ in the deficit identity and change variable to the volume parameter $s = m(t)$, in which the layer is exactly linear, $|\Omega \setminus E_t| = \omega_n - s$. The deficit bound becomes $\int_0^{\omega_n} \Phi(s) ds \leq C_n \delta_T$ for the transported integrand Φ ; Markov gives $|\{\Phi > \theta'\}| \leq C_n \delta_T / \theta'$, so a good $\hat{t}$ exists inside a near-boundary slab of s -length $O_\theta(\delta_T)$; intersect with the full-measure set of regular values (Sard). The weight w is two-sidedly $O(1)$ on that slab because $m \rightarrow \omega_n$ there. $\square \quad \square$

Common input, restated. At the good level, $|\nabla u|$ is close to a constant on $\Sigma = \Sigma_{\hat{t}}$ in L^2 (a small absolute constant, not a δ_T -power). Nothing pointwise, nothing geometric, is given for free. The two routes are two ways to use this.

2. WHAT STRATEGY A MUST DO, AND THE ROLE OF PART A

Strategy A wants the *geometric* conclusion that Σ is a small graph over a sphere. Your proposed mechanism has two steps.

(A1) *Self-quantification.* From the L^2 smallness of $|\nabla u| - \bar{f}$ plus Markov, off a set of small $\mathcal{H}^{n-1}$ -measure the quantity $|\nabla u|$ is pinched into $[\bar{f} - \kappa, \bar{f} + \kappa]$. Quantitatively, by Chebyshev on (1) (this is exactly Lemma 3.4 below), the set $\{|\nabla u| \leq s\} \cap \Sigma$ has measure $\lesssim s D_H / \bar{f}^2$.

(A2) *Poisson smoothing.* Pass to an interior level $\{u > t\}$, $t \geq C \delta_T$ with C large, and use *interior elliptic regularity*: the value of a solution at interior distance d from the boundary is, by the Poisson representation, an average of boundary data, so a boundary irregularity concentrated on a small set is *smoothed out* a definite distance inside. There is no dispersion; this is the standard fact that $\|\nabla^k(\text{harmonic ext})\|_{L^\infty(\{\text{dist} \geq d\})} \lesssim d^{-k-(n-1)/p} \|\text{data}\|_{L^p(\partial)}$.

The decisive question is the constant. If one estimates the interior from a *rough* $\partial\Omega$, the factor $d^{-k-(n-1)/p}$ blows up as $d \downarrow 0$, and the good level sits in a δ_T -thin shell, so $d \sim \delta_T \rightarrow 0$ and the estimate is vacuous. This is why the premise matters: **Strategy A presupposes Part A**, namely that a prior selection/regularization step has already produced a domain whose boundary is smooth with δ_T -uniform C^k bounds. *Given that premise*, global up-to-the-boundary Schauder makes $u \in C^{k+1}(\bar{\Omega})$ with δ_T -uniform norms; the interior constant no longer depends on d , the small bad set is genuinely smoothed on a slightly deeper level, and pointwise $|\nabla u| \approx \bar{f}$ together with the controlled geometry bootstraps the higher derivatives of the graph (an interpolation between small L^∞ and the fixed high norm). Under that premise the mechanism is sound; its single hard point is the premise itself, i.e. whether the selection/regularization genuinely delivers a δ_T -uniform smooth boundary. (That is precisely the unremoved compactness in the BDV selection principle.)

3. WHAT ROUTE B DOES INSTEAD

Route B refuses the geometric conclusion. It keeps everything *interior* and in L^2 , and it is Serrin's overdetermined problem handled by Weinberger's argument, made quantitative.

3.1. The Weinberger function. For the torsion function v of a domain D (so $-\Delta v = 1$, $v = 0$ on ∂D), set $P := |\nabla v|^2 + \frac{2}{n}v$. Because $\Delta v \equiv -1$ is constant,

$$\Delta P = \Delta |\nabla v|^2 + \frac{2}{n} \Delta v = 2|D^2 v|^2 + 2\nabla v \cdot \nabla(\Delta v) - \frac{2}{n} = 2\left(|D^2 v|^2 - \frac{1}{n}\right).$$

The pointwise Cauchy–Schwarz inequality $|D^2 v|^2 \geq (\text{tr } D^2 v)^2/n = (\Delta v)^2/n = 1/n$ gives $\Delta P \geq 0$; equality at a point holds iff $D^2 v = -\frac{1}{n}I$ there, i.e. iff v agrees to second order with a ball quadratic

$a - |x - x_0|^2/(2n)$. Writing the trace-free Hessian one checks the exact identity

$$(2) \quad \Delta P = 2 \left| (D^2 v + \tfrac{1}{n} I) \right|^2, \quad \left| (D^2 v + \tfrac{1}{n} I) \right|^2 = |D^2 v|^2 - \tfrac{1}{n} \geq 0,$$

so the *interior* quantity $\mathcal{R}(D) := \int_D |(D^2 v + \tfrac{1}{n} I)|^2$ measures non-roundness, vanishing iff D is a ball. Serrin/Weinberger rigidity is “ $\mathcal{R}(D) = 0 \Rightarrow D$ a ball”; Route B is its quantitative form, in two halves.

3.2. Half 1: the deficit dominates $\mathcal{R}$. We make the two classical identities explicit.

Rellich–Pohozaev. Test $-\Delta v = 1$ with $x \cdot \nabla v$. Using $v = 0$ (hence $\nabla v = v_\nu \nu$) on ∂D and the two divergence identities

$$\int_D \nabla v \cdot \nabla (x \cdot \nabla v) = \left(1 - \tfrac{n}{2}\right) \int_D |\nabla v|^2 + \tfrac{1}{2} \int_{\partial D} |\nabla v|^2 (x \cdot \nu), \quad \int_D 1 \cdot (x \cdot \nabla v) = -n \int_D v,$$

together with $\int_D |\nabla v|^2 = \int_D v =: T(D)$, one obtains after cancellation

$$(3) \quad \tfrac{1}{2} \int_{\partial D} |\nabla v|^2 (x \cdot \nu) d\mathcal{H}^{n-1} = \tfrac{n+2}{2} T(D).$$

Only the divergence theorem on a regular level is used — no boundary regularity of the original Ω .

Weinberger. Integrate (2). Pairing $\Delta P = 2|(D^2 v + \tfrac{1}{n} I)|^2$ against the Green function of D (a nonnegative weight available with no boundary regularity) converts the left side into $\int_D |(D^2 v + \tfrac{1}{n} I)|^2$ up to dimensional factors; the boundary terms produced by ΔP , using the boundary value $P = |\nabla v|^2$ on ∂D , are exactly the deviation of $|\nabla v|$ from its Serrin constant. Combined with (3) and the Saint–Venant gap $T(B_D) - T(D) \geq 0$ (a nonnegative boundary functional vanishing only on balls), one gets:

Proposition 3.1. *For a regular level domain D there is a purely dimensional $c(n) > 0$ with*

$$\delta_T(D) \geq c(n) |D|^{-1} \mathcal{R}(D),$$

no boundary regularity, no interior-ball or shape constant, additive over connected components.

The pointwise Newton inequality $|D^2 v|^2 \geq 1/n$, sharp only on the ball quadratic, is exactly what makes the boundary and interior expressions comparable and forces the relation to be *order two* (quadratic, the sharp Faber–Krahn exponent).

3.3. Half 2: $\mathcal{R}$ controls the asymmetry.

Lemma 3.2 (Hessian rigidity / quantitative Liouville). *Let D be open, $|D| = \omega_n$, $w \in H^2(D)$. There is a quadratic $q(x) = a - |x - x_0|^2/(2n)$ and a constant $K(D)$ with*

$$\int_D |\nabla w - \nabla q|^2 \leq K(D) \int_D |D^2 w + \tfrac{1}{n} I|^2.$$

$K(D)$ is a Poincaré-type constant of D .

Proposition 3.3. *With D a good level domain there is a dimensional $C(n)$ with $\mathcal{A}(D)^2 \leq C(n) |D|^{-1} \mathcal{R}(D)$.*

Mechanism. $\mathcal{R}(D)$ small means $D^2 v \approx -\tfrac{1}{n} I$ in L^2 ; by Lemma 3.2, v is H^2 -close to a ball quadratic q on the bulk; the superlevel set $\{v > \hat{t}\}$ is then close in measure to the ball $\{q > \hat{t}\}$ at the sharp quadratic rate (an $H^2 \hookrightarrow$ trace estimate turns the interior closeness into the squared L^2 size of the boundary deviation, which is $\mathcal{A}^2$). The only delicate point is that $K(D)$ in Lemma 3.2 could degrade where ∇v is small. This is controlled by the small-gradient lemma below: that set has small measure and is *excised*, paid for additively. $\square$

3.4. The role of D_H : measure-control, not graphing.

Lemma 3.4 (small-gradient set). *On $\Sigma = \Sigma_{\hat{t}}$, for every $s > 0$,*

$$\mathcal{H}^{n-1}(\{x \in \Sigma : |\nabla u(x)| \leq s\}) \leq \frac{4s}{\bar{f}^2} \int_{\Sigma} \frac{(|\nabla u| - \bar{f})^2}{|\nabla u|} d\mathcal{H}^{n-1}.$$

Proof. Where $|\nabla u| \leq s \leq \bar{f}/2$ one has $(|\nabla u| - \bar{f})^2 \geq \bar{f}^2/4$ and hence $(|\nabla u| - \bar{f})^2/|\nabla u| \geq \bar{f}^2/(4s)$; integrate (1)’s integrand over that set only and divide. $\square$

By Lemma 1.1 the right side is a small absolute constant, so the small-gradient set has small measure. *This is the only place D_H is used in Route B:* not to make Σ a graph, but to measure-bound the set where the Hessian-rigidity constant could degrade, so that it can be removed. For a disconnected good level one argues per component: a component carrying order-one asymmetry has order-one deficit by the qualitative Saint–Venant inequality (a threshold, by compactness), hence lies outside the small-deficit regime; so the per-component treatment loses nothing.

3.5. Assembly. Combining Propositions 3.3 and 3.1 on the good level domain, then the exact transfer $\delta_T(\text{level}) \leq 2\delta_T(\Omega)$ (the shifted torsion function is exactly the level domain’s) and the elementary superlevel transfer $\mathcal{A}(\Omega) \leq \mathcal{A}(\text{level}) + 2|\Omega \setminus \text{level}|/|\Omega|$ with the discarded layer $O(\delta_T)$, gives $\mathcal{A}(\Omega)^2 \leq C_*(n)\delta_T(\Omega)$, the squared collar being $O(\delta_T^2) \ll \delta_T$.

4. THE PRECISE CONTRAST

- Strategy A needs the bad set *smoothed away*: pushed toward measure zero and the good control made *uniform and pointwise on the whole boundary*, so that the level set is a controlled small graph. That is the (A1)+(A2) program; its soundness rests entirely on Part A making the up-to-boundary Schauder constants δ_T -uniform.
- Route B never needs the bad set to vanish or any pointwise boundary control. It needs only that the bad set has *small measure* (true by Lemma 3.4, with no propagation and no Part A), then *deletes* it (Lemma 3.2 on the bulk) and extracts the conclusion from the interior L^2 Hessian via (2). No graph, no boundary norm, no Poisson propagation.

So the smoothing-by-Poisson idea is the right tool *for Strategy A*; Route B’s design point is that, arguing in the interior through Weinberger, that step is simply not needed — the same L^2 near-Serrin input is consumed by excision instead of by propagation.

5. THE SERRIN DICTIONARY

Serrin’s theorem: if the torsion function of D has constant gradient on ∂D , then D is a ball. There are two quantitative “stabilities” of this statement, and the two routes are exactly the two of them:

- *Geometric/boundary Serrin stability* (Aftalion–Busca–Reichel, Magnanini–Poggesi, Brandolini–Nitsch–Salani–Trombetti type): $|\nabla v|$ near constant on $\partial D \Rightarrow \partial D$ close to a sphere *as a graph*. This needs boundary regularity and is what Strategy A invokes.
- *Interior/Weinberger rigidity stability*: the integrated trace-free Hessian $\mathcal{R}(D)$ is controlled by the deficit and controls the asymmetry, with no boundary statement at all. This is Route B.

Both descend from the same Weinberger P -function; they differ only in whether one reads the equality case off the *boundary* (needs regularity) or off the *interior* integral (2) (does not). That is the entire conceptual content of “Route B uses Serrin directly”.

6. A WORKED ILLUSTRATION: BALL PLUS THIN TENTACLE

Let $\Omega = B_1 \cup T$ with T a tube of length $L = O(1)$ and cross-radius $w \ll 1$, $|T| \simeq \kappa_n w^{n-1} L$. On the tube the transverse Dirichlet barrier $\phi(r) = (w^2 - r^2)/(2(n-1))$ caps u : $\sup_T u \simeq_n w^2$, independently of L . An energy expansion gives

$$\delta_T(\Omega) \simeq_n |T| = \kappa_n w^{n-1} L, \quad \mathcal{A}(\Omega) \simeq_n |T|,$$

and, since on the thin tube $D^2 u \approx -\frac{1}{n-1} I_{n-1}$ transversally (not ≈ 0), $|D^2 u + \frac{1}{n} I|^2 = \Theta_n(1)$ per unit volume there, so $\mathcal{R}(\Omega) \simeq_n |T|$. Hence Route B's chain reads

$$\mathcal{A}(\Omega)^2 \simeq |T|^2 \ll |T| \simeq \mathcal{R}(\Omega) \simeq \delta_T(\Omega),$$

so $\mathcal{A}^2 \lesssim \delta_T$ holds *with enormous room*: a tentacle is *not* a counterexample. By contrast the graph route must reach a tentacle-free level; that costs discarding a ball-shell of measure $\simeq_n w^2$ (the level $\{u > t\}$ for $t > \sup_T u \simeq w^2$ peels an annulus of thickness $\simeq w^2$ off the ball part), while the budget is $O(\delta_T) = O(w^{n-1} L)$. For $n \geq 4$, $w^2 \gg w^{n-1} L$: the graph route cannot escape the tentacle within an $O(\delta_T)$ layer, whereas Route B never tries to — it reads the conclusion off the interior $\mathcal{R}$, where the tentacle is harmless. (Note: the tube-level $D_H \simeq_n w^{n-3} L$ tends to 0 for $n \geq 4$, so such a tentacle does *satisfy* the near-Serrin hypothesis; this is exactly why the graph route is genuinely obstructed there and the interior route is not.)

7. HONEST STATUS

This note explains the *design* — why Route B avoids the Poisson-propagation step — not a proof. Two statements are load-bearing and not human-refereed:

Open point (Half 1). Proposition 3.1: that the Saint-Venant deficit dominates $|D|^{-1} \mathcal{R}(D)$ with a *purely dimensional* constant, valid on a regular level domain with no boundary regularity and stable under disconnection and thin tentacles. The exact tracking of the Green-function pairing constant is what must be checked.

Open point (Half 2). That the Poincaré constant $K(D)$ of Lemma 3.2 stays *dimensional* on the actual good level domain — possibly pinched, non-convex, multi-component — after excising the small-gradient set of Lemma 3.4. This is the single most important line to referee.

Symmetrically, Strategy A's single hard point is the Part A premise (a δ_T -uniform smooth boundary), which is the unremoved compactness of the BDV selection principle. Neither route is refereed; they are pursued in parallel because each has exactly one distinct analytic crux of comparable standing, and the apparent internal coherence of either is not, by itself, evidence — only external scrutiny is.